

MPC-MAP Assignment 2

Week 3: Motion Control

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Task 1 – Custom Map

A custom map `custom_1.txt` was created. The arena is 16×10 m with two internal walls. The goal is placed at $(14, 8)$. The entire arena is a GNSS-denied zone.

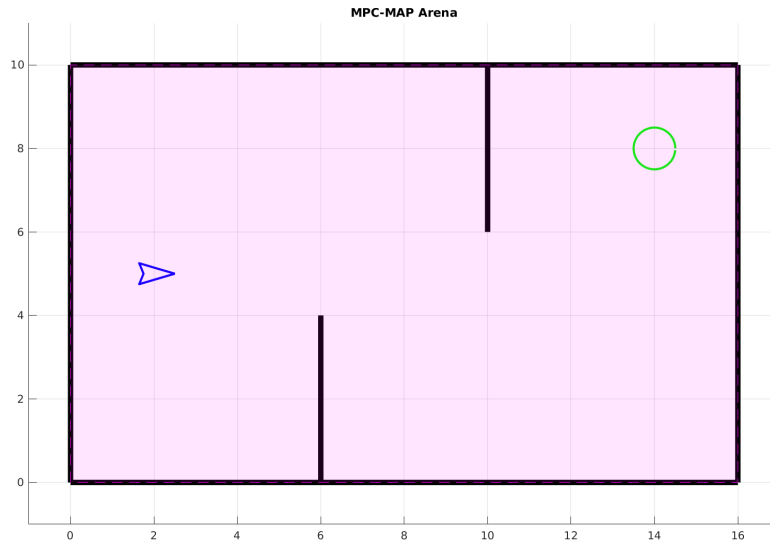


Figure 1: Custom map `custom_1` with two walls and the goal (green circle).

Task 2 – Sample Paths

Three test paths were defined as sequences of (x, y) waypoints generated in `student_workspace.m`:

- **Straight line** from $(2, 5)$ to $(14, 5)$, 50 waypoints.
- **Circular arc** with center $(8, 5)$ and radius 2 m, spanning a semicircle (60 waypoints).
- **Sine wave** from $x = 2$ to $x = 14$ with amplitude 1.5 m around $y = 7$ (80 waypoints).

The active path is selected by changing the `path_type` variable. Waypoints are stored in `public_vars.path` and displayed by the simulator automatically.

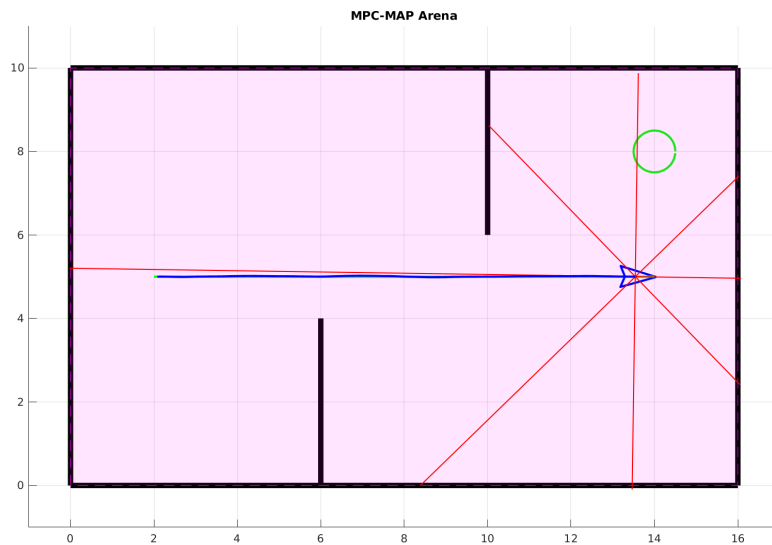


Figure 2: Straight-line path and the robot trajectory.

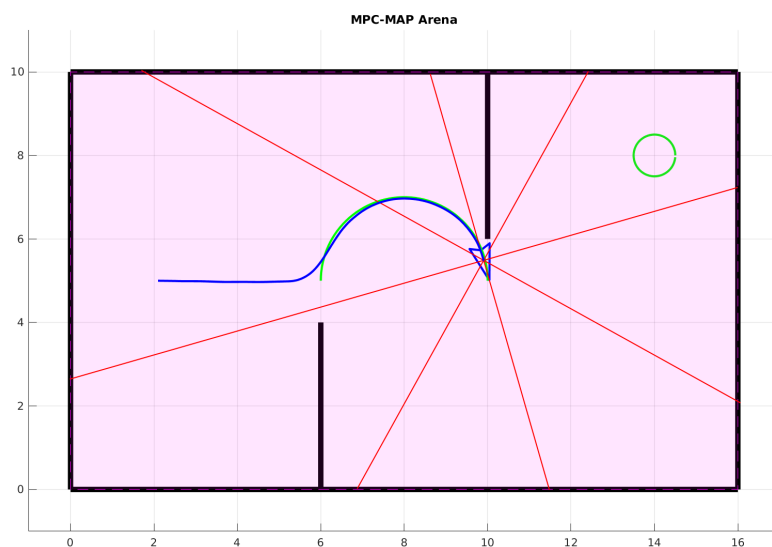


Figure 3: Circular-arc path and the robot trajectory.

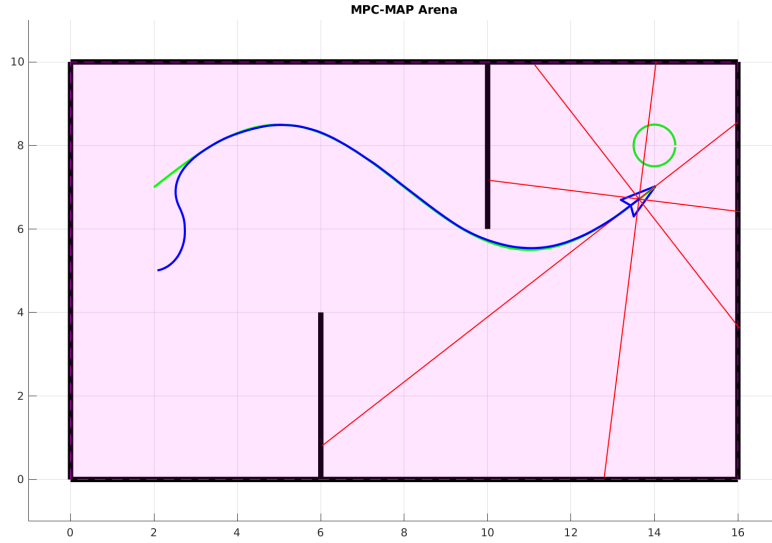


Figure 4: Sine-wave path and the robot trajectory.

Task 3 – Motion Control

A proportional path-following controller was implemented in `plan_motion.m`. The algorithm works in three steps:

1. `get_target` finds the next waypoint on the path using a lookahead distance of 0.8 m. It picks the first waypoint that is at least this far from the robot, searching forward from the closest point on the path.
2. The heading error e_θ between the robot orientation and the direction to the target is computed and wrapped to $[-\pi, \pi]$.
3. Forward speed v and angular rate ω are obtained with proportional gains:

$$v = K_v \cdot d, \quad \omega = K_\omega \cdot e_\theta$$

where d is the distance to the target. These are then converted to wheel velocities for the differential drive:

$$v_R = v + \frac{\omega L}{2}, \quad v_L = v - \frac{\omega L}{2}$$

where $L = 0.2$ m (wheel distance).

The robot pose is read from `read_only_vars.mocap_pose` (motion capture).